

HOMEWORK 4

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 10/02/2023

Problem 1. Under the assumption that no pivoting (i.e., no row interchanging) is required, implement Gaussian elimination as `ge_lu(A)`, taking A as an input and returning L and U stored in A (submit your code with this problem). Then implement an `lusolve(A,b)` command that takes the output of your `ge_lu` algorithm and solves $Ax = b$ in $\mathcal{O}(n^2)$ floating point operations.

Problem 2. A matrix A is called *tridiagonal* if it has nonzero entries on only the main diagonal and the diagonal row one above/below the main diagonal; i.e., if it is of the form

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & 0 & \dots & 0 \\ a_{2,1} & a_{2,2} & \ddots & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & a_{n-1,n-2} & a_{n-1,n-1} & a_{n-1,n} \\ 0 & \dots & 0 & a_{n,n-1} & a_{n,n} \end{bmatrix}$$

Write an algorithm in pseudocode that takes as input a tridiagonal matrix $A \in \mathbb{R}^{n \times n}$ and a vector $b \in \mathbb{R}^n$ and solves $Ax = b$ in $cn + \mathcal{O}(1)$ floating point operations for some constant c . If you employ Gaussian elimination, you may assume that no pivoting (i.e., no row interchanging) is required.

Problem 3. The Gauss-Jordan method used to solve the prototype linear system can be described as follows. Augment A by the right-hand-side vector b and proceed as in Gaussian elimination, except use the pivot element $a_{kk}^{(k-1)}$ to eliminate not only $a_{ik}^{(k-1)}$ for $i = k + 1, \dots, n$ but also the elements $a_{ik}^{(k-1)}$ for $i = 1, \dots, k - 1$, i.e., all elements in the k^{th} column other than the pivot. Upon reducing $\left[A \mid B \right]$ into

$$\left[\begin{array}{cccc|c} a_{11}^{(n-1)} & 0 & \dots & 0 & b_1^{(n-1)} \\ 0 & a_{22}^{(n-1)} & \dots & \vdots & b_2^{(n-1)} \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & a_{nn}^{(n-1)} & b_n^{(n-1)} \end{array} \right]$$

the solution is obtained by setting $x_k = \frac{b_k^{(n-1)}}{a_{kk}^{(n-1)}}$ for $k = 1, \dots, n$. This procedure circumvents the backward substitution part necessary for the Gaussian elimination algorithm.

- Write a pseudocode for this Gauss-Jordan procedure using, e.g., the same format as for the one appearing in Section 5.2 for Gaussian elimination. You may assume that no pivoting (i.e., no row interchanging) is required.
- Show that the Gauss-Jordan method requires $n^3 + \mathcal{O}(n^2)$ floating point operations for one right-hand-side vector b (this is roughly 50% more than what's needed for Gaussian elimination).